FinTrust Digital Bank - Data Analytics Track

Week 4, Part E: Final Validation

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Eight findings were validated by recomputing them from the raw CSV files and, where a comparison is involved, testing whether the difference is larger than random variation (chi-square, Kruskal-Wallis, binomial test, 95% confidence interval; significance threshold $p < 0.05$). All checks are in the script *FinTrust_Week4_Validation_Checks.py*, with its output saved in *FinTrust_Week4_Validation_Output.txt*. Status key: **Confirmed** = held as stated, **Refined** = held but wording sharpened, **Revised** = conclusion changed.

Validation 1: Dashboard KPIs are accurate; SQL totals reconciled | Status: Refined	
Finding	The six KPI cards (1,500 customers, 12,000 transactions, NGN 560.48M total value, NGN 46.71K average, 90.47% success rate, 19.60% risk-review rate) are correct, and SQL agrees with them.
Validation Method	Recomputed every KPI from the raw CSVs in Python, then compared with the Power BI cards and with SQL Server (SUM, AVG and COUNT of Amount_NGN).
Result	Python and Power BI match: 1,500 customers, 12,000 transactions, NGN 560,477,354.85 total, NGN 46,706.45 average, 90.47% success, 19.60% risk rate. SQL Server initially returned 554,157,109.34 (average 46,677.65). The gap of NGN 6,320,245.51 equals the 128 transactions with whole-number amounts that were imported as NULL; the CSV has no missing amounts. Counts and rates in SQL matched throughout.
Conclusion	Dashboard KPIs confirmed accurate. The SQL import was repaired (restore 128 amounts, DECIMAL(18,2)). After the repair SQL returns NGN 560,477,355.91 with 0 NULL amounts, within NGN 1.06 of the CSV total (rounding left over from the original float import).

Validation 2: International transactions carry about double the risk-review rate | Status: Confirmed	
Finding	International risk-review rate is about twice the domestic rate.
Validation Method	Recomputed rates from raw data; chi-square test of international status against risk flag.
Result	International 36.9% (177 of 480) vs domestic 18.9% (2,175 of 11,520). Chi-square p = 3.96e-22.
Conclusion	Confirmed with very strong statistical support. This is the most reliable risk finding in the project.

Validation 3: Customer segment does not differentiate risk or spend per customer | Status: Refined	
Finding	Segment is a weak differentiator of risk. Everyday dominates totals.
Validation Method	Chi-square of segment against risk flag; Kruskal-Wallis test of per-customer spend across segments; per-customer spend by segment.
Result	Risk rate 19.3% to 20.3% (p = 0.75). Spend per customer NGN 365K to 388K (Kruskal-Wallis p = 0.46). Everyday has 47.4% of customers and 46.6% of value.
Conclusion	Confirmed for risk. Refined for value: Everyday leads in totals because of headcount, not higher spend per customer.

Validation 4: Transaction value predicts risk flagging | Status: Confirmed	
Finding	Larger transactions are flagged for review far more often.
Validation Method	Flag rate by five value ranges with chi-square test; top 1% (by value) against the other 99% with chi-square test.
Result	17.7%, 17.8%, 18.9%, 25.5%, 40.9% from lowest to highest range (p = 1.6e-40). Top 1% flagged 35.83% vs 19.44% (p = 1.2e-05).
Conclusion	Confirmed. Review activity rises clearly with value, above NGN 100K in particular.

Validation 5: Success rate by value range | Status: Refined	
Finding	Earlier wording: transaction value does not affect failure.
Validation Method	Success rate by value range with chi-square test.
Result	Success 91.0%, 89.2%, 89.3%, 91.3%, 92.3% (p = 0.011). The difference is detectable but small (about 3 points) and the highest-value range has the highest success rate.
Conclusion	Refined: value is not a meaningful driver of failure. The earlier wording 'no effect' is replaced by 'small, non-directional difference'.

Validation 6: Failures and reversals are concentrated in USSD | Status: Revised	
Finding	Weeks 2-3 conclusion: failure and reversal are concentrated in USSD.
Validation Method	Chi-square of channel against failed/reversed status; binomial test of the highest channel-and-type cells against the overall rate (7.97%).
Result	Channel rates: ATM 6.9%, Web 7.1%, POS 7.7%, Mobile App 8.6%, USSD 8.7% (p = 0.070, not significant). USSD Cash Withdrawal 10.0% (n = 100, p = 0.46) and USSD Deposit 10.0% (n = 130, p = 0.42) are within normal variation.
Conclusion	Revised: failure is platform-wide at about 8%. No channel is statistically worse. The USSD recommendation is withdrawn.

Validation 7: ATM + International is the highest-risk combination | Status: Revised	
Finding	Weeks 3 conclusion: ATM + International (46.0%) is the sharpest risk hotspot.
Validation Method	95% confidence interval for the ATM + International rate; chi-square of channel against risk within international transactions.
Result	n = 63, rate 46.0%, 95% CI 33.7% to 58.3%. Differences between channels within international are not significant (p = 0.27).
Conclusion	Revised: the supported finding is that international status as a whole drives risk. ATM is a lead to monitor, not a proven hotspot.

Validation 8: The February dip is a genuine fluctuation | Status: Revised	
Finding	Weeks 2-3 conclusion: February's lower value may be a real fluctuation.
Validation Method	Compared monthly value and transaction counts per calendar day.
Result	Days: 31, 28, 31. Transactions per day 133.3, 133.4, 133.3. Value per day NGN 6.08M, 6.28M, 6.33M.
Conclusion	Revised: the dip is a calendar effect (28 days). Daily activity is flat to slightly rising.

Summary of findings that changed after validation

Finding	Before validation	After validation
SQL value totals	Matched across SQL, Python and Power BI	SQL was 1.1% low (128 amounts imported as NULL); repaired; counts and rates were unaffected
Failure concentration	Concentrated in USSD	Platform-wide, about 8%; no channel significantly worse
Highest-risk combination	ATM + International (46.0%) to be auto-flagged	International overall is the driver (36.9% vs 18.9%); ATM is directional only (n = 63)
February dip	Possibly a genuine fluctuation	Calendar effect; daily value slightly rising
Value and failure	Value has no effect	Small, non-directional difference (about 3 points)
Everyday segment	Everyday is the core of the business	True in totals, driven by headcount; spend per customer is similar across segments

What this validation can and cannot show. It shows which patterns in this dataset are larger than chance. It cannot show causes (for example, why a transaction failed) because the data has no failure-reason field. The data covers three months and is synthetic. International transactions number 480, so breakdowns within them have wide uncertainty. Risk_Review_Flag records transactions sent for review, not confirmed fraud.